

Phishing Email Detector: Model Efficiency Benchmark

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Executive Summary

This benchmark compares two approaches for phishing email classification on a dataset of 175,082 emails. The standard TF-IDF + Logistic Regression/Random Forest pipeline is compared against a fast Hashing Vectorizer + SGD Classifier pipeline. Results demonstrate significant efficiency gains while maintaining competitive accuracy.

Performance Metrics

Metric	TF-IDF + LR/RF	Hashing + SGD	Improvement
Training Time (s)	1247.5	186.3	6.7x faster
Inference Time (ms)	12.4	2.8	4.4x faster
Memory Usage (MB)	542	128	4.2x lower
Accuracy	0.9284	0.9106	-1.8%
F1 Score (Macro)	0.9231	0.9034	-2.2%
Precision	0.9512	0.9421	-0.9%
Recall	0.9156	0.8934	-2.5%

Key Findings

- **Speed:** Hashing+SGD is 6.7x faster to train and 4.4x faster for inference.
- **Memory:** 4.2x reduction in memory footprint due to fixed feature dimension.
- **Accuracy Trade-off:** Minor accuracy loss (1.8%) is acceptable for operational benefits.
- **Scalability:** Hashing Vectorizer scales linearly; TF-IDF requires vocabulary fitting.
- **Real-time Performance:** 2.8ms inference enables real-time email scanning.

Recommendation

Deploy the Hashing+SGD model for production inference to achieve sub-3ms latency per email. Use the TF-IDF model for periodic retraining tasks where accuracy is prioritized over speed. Implement ensemble scoring for critical decisions: use SGD for fast pre-filtering, TF-IDF for detailed analysis.

Resources & References

1. HashingVectorizer vs TfidfVectorizer:
https://scikit-learn.org/stable/modules/feature_extraction.html#hashing-trick
2. SGDClassifier for Online Learning:
https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.SGDClassifier.html
3. Email Phishing Detection Benchmarks: <https://arxiv.org/abs/1910.07322>